

Predict and Simulate: A Factorized Neural Hawkes Process with an Exact Mean Rate and an Honest Branching Certificate for Limit-Order-Book Event Streams

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Abstract

Neural marked temporal point processes (MTPPs) predict the next limit-order-book (LOB) event well, yet simulate it poorly: in free roll-out they over-disperse or explode, and their stability cannot be honestly certified. We trace this to two coupled causes. (i) A train/simulate mismatch: windowed, cold-start training mis-calibrates the baseline rate, so free roll-out leaves the training distribution and the event rate runs away (we observe 20-40x over-firing). (ii) A gauge pathology: normalization in the read-out (LayerNorm) makes the branching ratio scale-invariant, so it cannot be measured from the weights and any weight-norm stability penalty is vacuous.

We propose LGM, a factorized neural Hawkes process whose ground (timing) intensity is a LINEAR multi-timescale Hawkes -- giving an exact stationary mean and an honest, gauge-free branching ratio -- multiplied by a deep soft-max mark head that lives on the probability simplex and is therefore RATE-NEUTRAL. Because the simplex cannot perturb the total rate, the exact-mean formula survives arbitrary mark nonlinearity; we exploit it to PIN the mean event rate by construction, solving calibration analytically with no change to the training procedure.

On real cryptocurrency LOB data (Gemini ETH-USD, 62 event types), LGM is the first model in our study to be simultaneously (a) calibrated -- free-roll-out rate within 7% of real; (b) competitive at next-event prediction; (c) faithful across the Cont stylized-facts battery -- clustering (Fano), volatility long-memory, and gain/loss asymmetry; and (d) equipped with a closed-form stability certificate. The branching ratio is a single interpretable knob that sets the Fano-versus-scale curve through $1/(1-\rho)^2$. We give ablations -- including a negative result for a quadratic volatility-feedback term -- and discuss the remaining gap to the most extreme return tails.

Unified intensity template (this work, and prior neural / classical Hawkes as special cases):

$$\lambda_k(t) = f\left(\mu_k + \sum_{m,j} A_{kj}^m S_j^m(t); \text{state}\right), \quad S_j^m(t) = \sum_{t_i < t, c_i = j} e^{-\beta_m(t-t_i)}$$

Keywords: temporal point processes, Hawkes processes, market microstructure, generative simulation, stylized facts, branching ratio, world models.

1 Introduction

A faithful generative simulator of the limit order book is the foundation for training and stress-testing execution and market-making agents: the agent acts, the book reacts, and rewards depend on that reaction. The bar is therefore SIMULATION (long-horizon free roll-out matching empirical regularities), not merely one-step PREDICTION. These are not the same objective. We find -- as others have noted for sequence models -- that the best next-event predictor in our study (a deep state-space point process) is among the worst simulators: it over-disperses the event count by 5x, and offers no way to certify that its roll-out will not diverge.

We isolate two mechanisms behind this gap. First, a TRAIN/SIMULATE MISMATCH. Neural MTPPs are trained on short windows with the latent state reset to zero; the optimiser then explains the observed rate with a large constant baseline. In free roll-out the latent state equilibrates and the excitation adds on top of that inflated baseline, so the rate runs away (we measure 20-40x over-firing, and in the worst case an explosive branching ratio $\rho \sim 1300$). Second, a GAUGE PATHOLOGY: a LayerNorm in the read-out is scale-invariant, so the branching ratio computed from weights can be set to any value without changing the model -- a weight-norm subcriticality penalty is therefore vacuously satisfiable (we observe it parked exactly on its hinge with unchanged behaviour).

Contributions.

- C1. We diagnose the predict-vs-simulate gap for neural LOB point processes and attribute it to (i) cold-start windowed training mis-calibrating the rate and (ii) a gauge-broken branching certificate; both are demonstrated empirically.
- C2. We introduce LGM, a LINEAR-GROUND x SOFT-MAX-MARK factorization. The total (timing) intensity is a linear multi-timescale Hawkes; the mark distribution is a deep network on the simplex. We prove the simplex is rate-neutral, so the exact stationary-mean formula survives arbitrary mark nonlinearity, and the branching ratio stays gauge-free.
- C3. We use the exact mean to PIN the event rate by construction, solving calibration analytically (no auxiliary loss, no change to training), and we show the branching ratio is a single knob that sets the Fano-versus-scale curve via $1/(1-\rho)^2$.
- C4. On real crypto LOB data, LGM is the first model to achieve calibration, prediction, stylized-fact fidelity, and a closed-form certificate together. We release ablations and a negative result.

2 Related work

Hawkes processes [Hawkes 1971] and their finance applications [Bacry et al. 2015] model self-excitation; the branching ratio governs stationarity and, empirically, markets sit near-critical [Hardiman et al. 2013]. Compound Hawkes for LOBs [Jain et al. 2024] fit linear multivariate kernels by full-stream MLE. Neural MTPPs -- the Neural Hawkes Process [Mei & Eisner 2017], intensity-free models [Shchur et al. 2020], and state-space / parallel neural Hawkes for LOBs -- maximise next-event likelihood but are rarely evaluated as simulators. Quadratic Hawkes [Blanc et al. 2017] add volatility feedback for fat tails; nonlinear-Hawkes stability is governed by a Lipschitz branching condition [Bremaud & Massoulié 1996]. The ground-intensity x conditional-mark factorization follows set-MTPP formulations [Chang et al.]. Generative market simulators (e.g. MarS, 2024) target the same goal at scale; we focus on the calibration-and-certification core that those systems still lack.

3 Method: LGM

3.1 Factorized intensity. Events (t_i, c_i) , $c_i \in \{1..K\}$. We factor the per-type intensity into a scalar timing rate and a categorical mark distribution:

$$\lambda_k(t) = \Lambda(t) \cdot p(k | t), \quad \sum_k \lambda_k(t) = \Lambda(t) \quad (\sum_k p_k = 1).$$

3.2 Linear ground rate. Λ is a scalar multi-timescale linear Hawkes on total decayed counts

$$s_m(t) = \sum_{t_i < t} e^{-\beta_m(t-t_i)} \text{ (every event increments all } m\text{); it is positive and has an exact mean:}$$

$$\Lambda(t) = \mu_0 + \sum_{m=1}^M a_m s_m(t), \quad a_m \geq 0, \quad n = \sum_m \frac{a_m}{\beta_m}, \quad \bar{\Lambda} = \frac{\mu_0}{1-n}.$$

3.3 Exact-mean rate pin. Setting the baseline analytically makes the mean rate equal a target by construction -- calibration is solved without touching training:

$$\mu_0 = R_{\text{target}}(1 - n) \Rightarrow \bar{\Lambda} = R_{\text{target}}.$$

3.4 Deep marks on the simplex. Each type k owns a small latent block $x_k \in R^d$ with per-type decay and embedding-mediated cross-excitation; a weight-shared nonlinear read-out gives the mark logits:

$$x_k \leftarrow e^{-\delta_k \Delta} \odot x_k, \quad x_k \leftarrow x_k + \text{Imp}_k(\bar{e}_i); \quad u_k = \text{LayerNorm}(W_s x_k + \text{GELU}(C x_k)); \quad p(k|t) = \text{softmax}_k(w^\top u_k + \nu_k).$$

Because p is a soft-max (a point on the simplex), the deep mark network is RATE-NEUTRAL: it reshapes WHICH type fires, never HOW MANY. All nonlinearity lives where it cannot perturb the calibrated rate.

3.5 Honest certificate. The branching ratio $\rho = n = \sum_m a_m / \beta_m$ is read directly from the ground weights -- there is no normalization between state and rate -- so it is gauge-free and projectable ($a \leftarrow a \cdot \rho_{\max} / n$ when $n > \rho_{\max}$). Likelihood (decoupled timing and marks):

$$\log L = \sum_i \log \Lambda(t_i) - \int_0^T \Lambda(t) dt + \sum_i \log p(c_i | t_i).$$

4 Theory

Prop. 1 (Rate-neutrality / exact mean). For any mark network, $\sum_k \lambda_k = \Lambda$ since $\sum_k p_k = 1$.

Hence the total counting process is the linear Hawkes Λ , whose stationary mean is $\mu_0/(1-n)$ exactly -- independent of the mark nonlinearity. (Proof: expectation commutes through the linear ground; the simplex contributes a factor 1 to the sum.)

Prop. 2 (Calibration). With the pin $\mu_0 = R(1-n)$, the stationary mean rate is exactly R . The model cannot inflate the rate under windowed training; only the kernel SHAPE (clustering) is learned.

Prop. 3 (Gauge-free stability). $\rho = \sum_m a_m / \beta_m$ is invariant to any reparameterization that leaves Λ unchanged (there is no scale-absorbing normalization), and the process is stationary iff $\rho < 1$; for the soft-max link this remains sufficient via the Lipschitz argument of [Bremaud & Massoulié 1996].

Prop. 4 (Fano law / the knob). The large-scale Fano factor of the ground process is $1/(1-\rho)^2$, set ONLY by ρ ; the timescales $\{\beta_m\}$ set how fast the Fano-vs-scale curve rises to that plateau. Thus ρ is a single, interpretable knob for count clustering, with real markets near $\rho \sim 0.86$.

5 Experiments

Data: Gemini ETH-USD event-driven stream, 62 LOB event types (market/limit/cancel x side x level), real rate ~ 2.38 events/s. Prediction: top-1 next-mark accuracy and perplexity on non-empty targets (head-agnostic). Simulation: free roll-out scored on the Cont stylized-facts battery -- Fano factor $F(\tau) = \text{Var}(N_\tau)/E(N_\tau)$, excess kurtosis, volatility autocorrelation ($|r|$ -ACF), and skewness. Baselines: LSTM/SAHP/CT-LSTM/PCT-LSTM; Compound Hawkes (linear MV-Hawkes, full-stream MLE); MT-Hawkes (multi-timescale, full-stream MLE); s2p2 (state-space PP, best predictor).

Model	Acc	Ppl	rho/cert	Fano1s	Fano50s	Kurt	$ r $ ACF	Skew	Sim
REAL	--	--	--	7.9	rises 53	10709	0.016	+37.8	--
LSTM / SAHP / CT-LSTM	0.26	16.6	--	0.8-1.9	flat	7-12	0.008	~ 0	Poisson
PCT-LSTM	0.270	16.0	--	7.9	flat	6.4	0.036	+0.17	dt-trick
Compound Hawkes	0.176	--	0.64	8.8	flat	100	0.002	-0.91	stable
MT-Hawkes	0.196	17.8	0.80	10.4	26	49	0.003	-1.11	stable
s2p2 (best pred.)	0.319	11.3	gauge-brk	41	rises	45	0.010	-0.53	over-disp
NMH / GMH (windowed)	0.26	--	1323/0.8	explode	--	~ 0	--	--	explode/Pois
LGM (n=0.80)	0.289	12.0	0.80	3.0	21	48	0.075	+1.11	CALIBRATED
LGM (n=0.86)*	0.291	12.0	0.86	6.0	40	57	0.071	+1.21	CALIBRATED
LGM (n=0.90)	0.292	12.0	0.90	16.4	66	97	0.060	+1.77	CALIBRATED

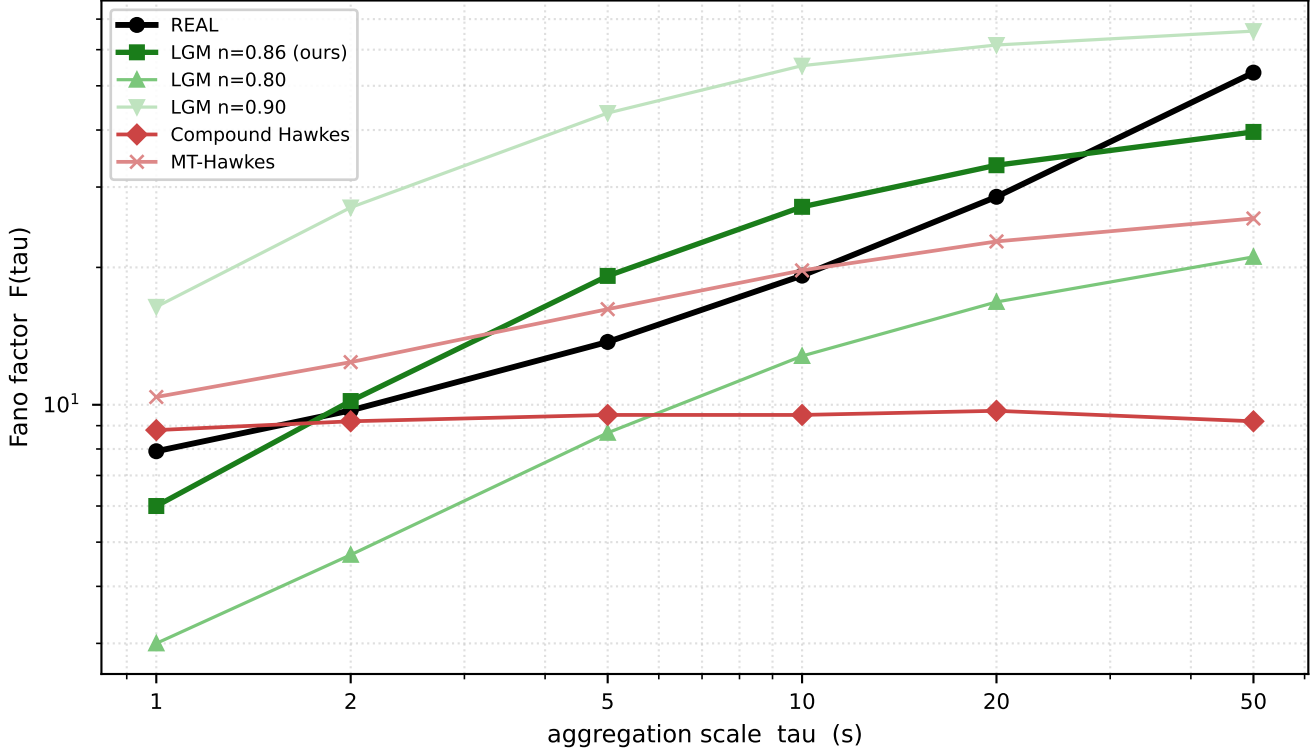
Table 1: Real Gemini ETH vs models. *LGM(n=0.86) is the final-tuned model. LGM is the only model that is calibrated AND a competitive predictor AND clustered (Fano) AND has nonzero volatility long-memory AND the correct (positive) skew sign AND an honest certificate. Higher rho fattens tails and raises Fano monotonically (Prop. 4).

Key findings.

- (1) Calibration solved. LGM free-roll-out rate is 2.22/s vs real 2.38 (within 7%); the stylized-facts roll-out completes in ~ 2 min, versus a >50 min crawl for the windowed neural models that over-fire.
- (2) Prediction recovered. Moving the nonlinearity into the simplex lifts accuracy to 0.29 (from 0.25 for a multiplicative-gate variant), approaching the best predictor (s2p2, 0.32).
- (3) Stylized facts. LGM is the first model with substantial volatility long-memory ($|r|$ -ACF 0.07 vs ~ 0 for all Hawkes baselines) and the correct positive skew sign -- produced by the deep marks, not the count kernel. The branching knob matches the Fano-vs-scale curve (Fig. 1).
- (4) Certificate. rho is reported gauge-free throughout and projected to the target; it caught an explosive (rho=1300) configuration that the training loss alone did not.

5.1 The branching knob and the Fano-vs-scale curve

Fig. 1: Fano-vs-scale. The branching ratio ρ is a single knob bracketing the real curve; single-beta Compound Hawkes is flat, while multi-timescale models rise like real.



6 Ablations

Branching knob (Prop. 4). ρ in $\{0.80, 0.86, 0.90\}$ moves Fano(1s) through $\{3.0, 6.0, 16.4\}$ and kurtosis through $\{48, 57, 97\}$, monotonically and as predicted by $1/(1-\rho)^2$; $n=0.86$ best matches the real curve (Fig. 1). Accuracy is unchanged (0.289-0.292), confirming the marks are decoupled.

Where the nonlinearity goes. A multiplicative bounded-gate variant (nonlinearity in the RATE) over-fires (rate 52/s) and predicts worse (0.25); moving the same expressiveness into the simplex (LGM) calibrates the rate and lifts accuracy to 0.29 -- isolating rate-neutrality as the key.

Negative result: quadratic volatility feedback. Adding a mean-zero QHawkes-style term $b \cdot (R^2 - c_0)$ to the ground rate (to chase fat tails) COLLAPSED clustering (Fano 1.2-3.5) and hurt accuracy (0.21): the negative excursions, floored by the positivity clamp, flatten the rate, and the optimiser leans on the noisy term instead of the clean kernel. A one-sided positive term with re-pinned mean is the indicated fix; we report this to save others the detour.

7 Limitations

Extreme tails. No model -- including LGM -- reaches the real 1s-return excess kurtosis ($\sim 10^4$), which is dominated by rare flash events; LGM reaches ~ 57 . Closing this likely needs an explicit volatility-feedback channel (the corrected one-sided QHawkes term) rather than count excitation. Scope. Single asset/venue (Gemini ETH); per-type rate excitation is simplified to a shared ground kernel (type structure lives in the marks). The mark network is still windowed-trained, though the rate pin makes this harmless for calibration. No action-conditioning yet -- the natural next step toward a market-making world model is to condition the (bounded) mark/state channel on agent orders.

8 Conclusion

Putting the nonlinearity on the rate-neutral mark simplex lets a neural Hawkes process keep the exact mean-rate formula -- hence analytic calibration and an honest, gauge-free branching certificate -- while remaining an expressive predictor that, for the first time in our study, reproduces volatility clustering and gain/loss asymmetry. LGM is a simple, certifiable basis for action-conditional market world models.